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UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-29 & MVD-30 20kV Centralized Panel Room WICA SCHEMATIC DIAGRAM SBC-W OUTGOING H09,H10,H11																																																																										
<table><tr><td></td><td></td><td></td><td></td><td>Date Prep.</td><td>15.11.2018</td><td colspan="2">PROJECT TITLE</td><td rowspan="2">ABB</td><td rowspan="2">EPMV</td><td rowspan="3">COVER SHEET</td><td>Scale</td><td>NTS</td><td colspan="2">= H09,H10,H11</td></tr><tr><td></td><td></td><td></td><td></td><td>Prepared</td><td>YR</td><td colspan="2">MV Switchgear for MNA Serang Plant</td><td colspan="2">MVD-29 & MVD-30 20kV Centralized Panel Room WICA</td><td>Customer</td><td>PT Multimas Nabati Asahan</td><td colspan="2">& 305</td></tr><tr><td>00</td><td>Issued for Approval</td><td>17.01.19</td><td>YR</td><td>Checked</td><td>AW</td><td colspan="2"></td><td colspan="2">PT. ABB Sakti Industri</td><td>SCHMATIC DIAGRAM</td><td>Drawing Nbr.</td><td>18.042-MNA-305.10</td><td colspan="2">Sh. 1</td></tr><tr><td>Ind.</td><td>Revision</td><td>Date</td><td>Name</td><td>Approved</td><td>AI</td><td colspan="2">Based on: BE318042</td><td colspan="2"></td><td>SBC-W OUTGOING</td><td>Customer Doc. Nbr.</td><td></td><td colspan="2">Cont. 27</td></tr></table>																				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	COVER SHEET	Scale	NTS	= H09,H10,H11						Prepared	YR	MV Switchgear for MNA Serang Plant		MVD-29 & MVD-30 20kV Centralized Panel Room WICA		Customer	PT Multimas Nabati Asahan	& 305		00	Issued for Approval	17.01.19	YR	Checked	AW			PT. ABB Sakti Industri		SCHMATIC DIAGRAM	Drawing Nbr.	18.042-MNA-305.10	Sh. 1		Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				SBC-W OUTGOING	Customer Doc. Nbr.		Cont. 27	
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==MVD-29_MVD-30=H09,H10,H11&305/4	REFERENCE DESIGNATIONS	00		
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==MVD-29_MVD-30=H09,H10,H11&305/6	REFERENCE DESIGNATIONS	00		
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			Prepared	YR				Customer	PT Multimas Nabati Asahan	& 305		
00	Issued for Approval	17.01.19	YR	Checked				AW	Drawing Nbr.	18.042-MNA-305.10	Sh. 2	
Ind.	Revision	Date	Name	Approved				AI	Based on:	BE318042	PT. ABB Sakti Industri	SCHEMATIC DIAGRAM SBC-W OUTGOING

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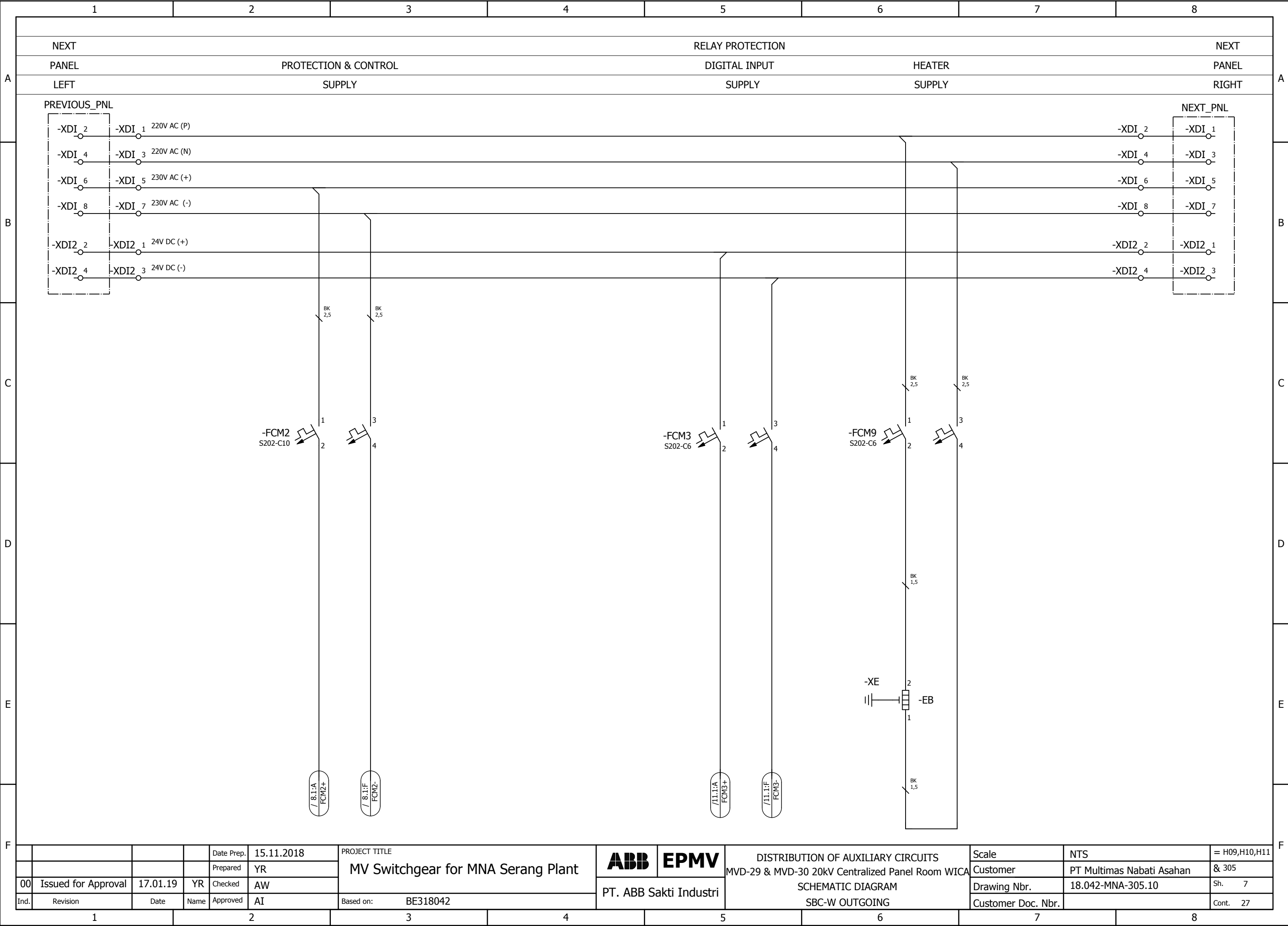
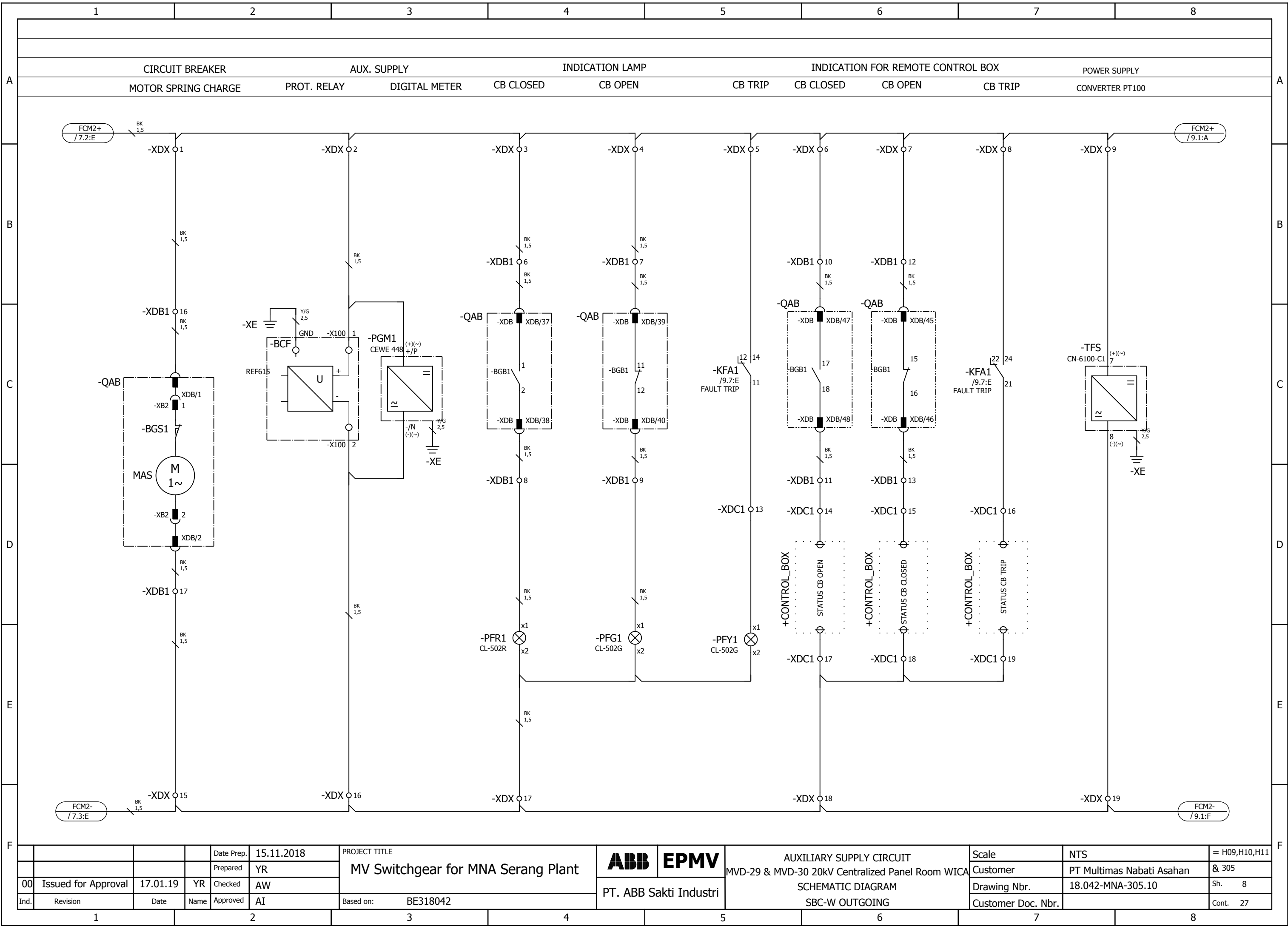


ABB **EPMV**

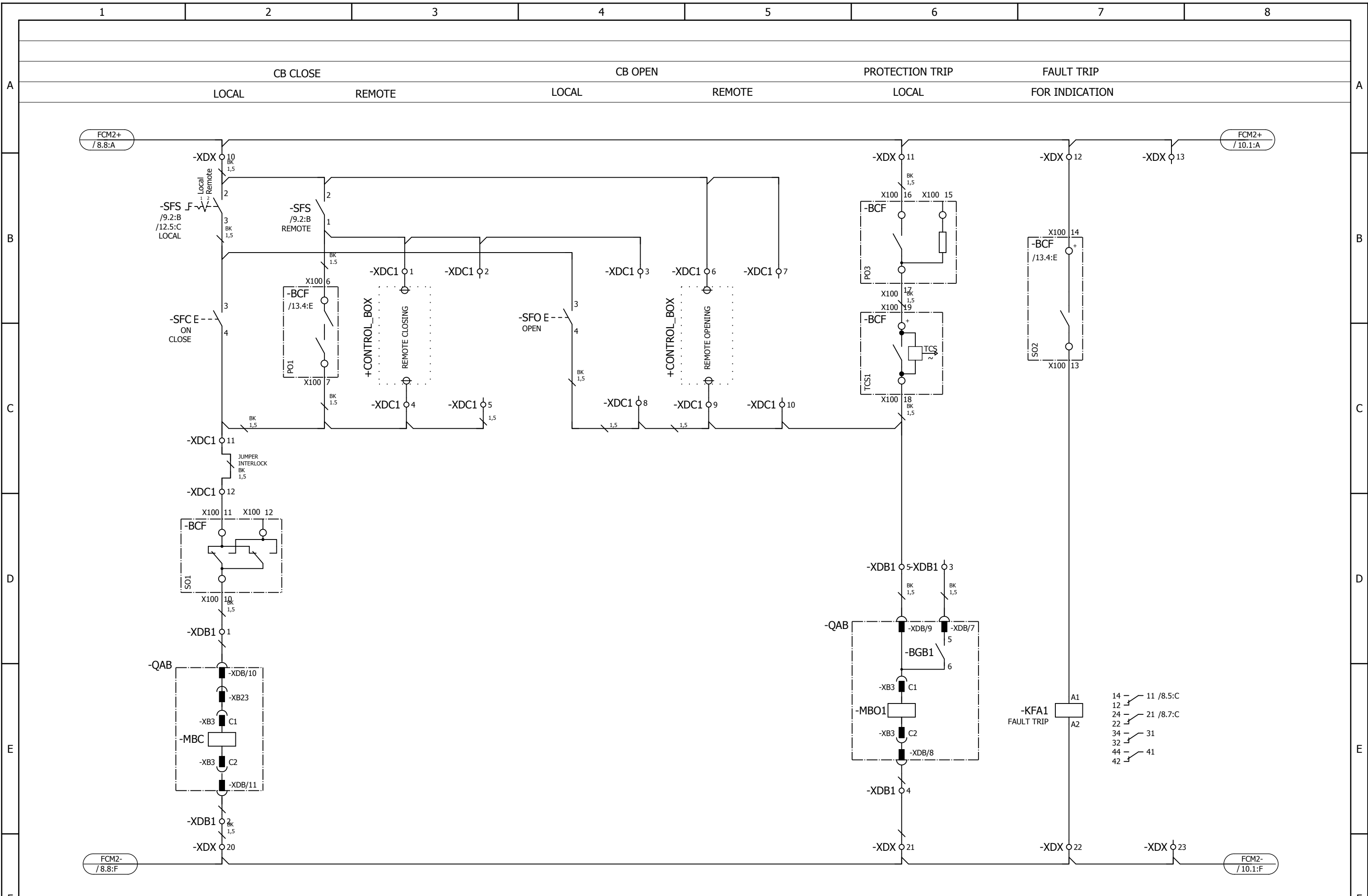
PT. ABB Sakti Industri

DISTRIBUTION OF AUXILIARY CIRCUITS
MVD-29 & MVD-30 20kV Centralized Panel Room WICA
SCHEMATIC DIAGRAM
SBC-W OUTGOING

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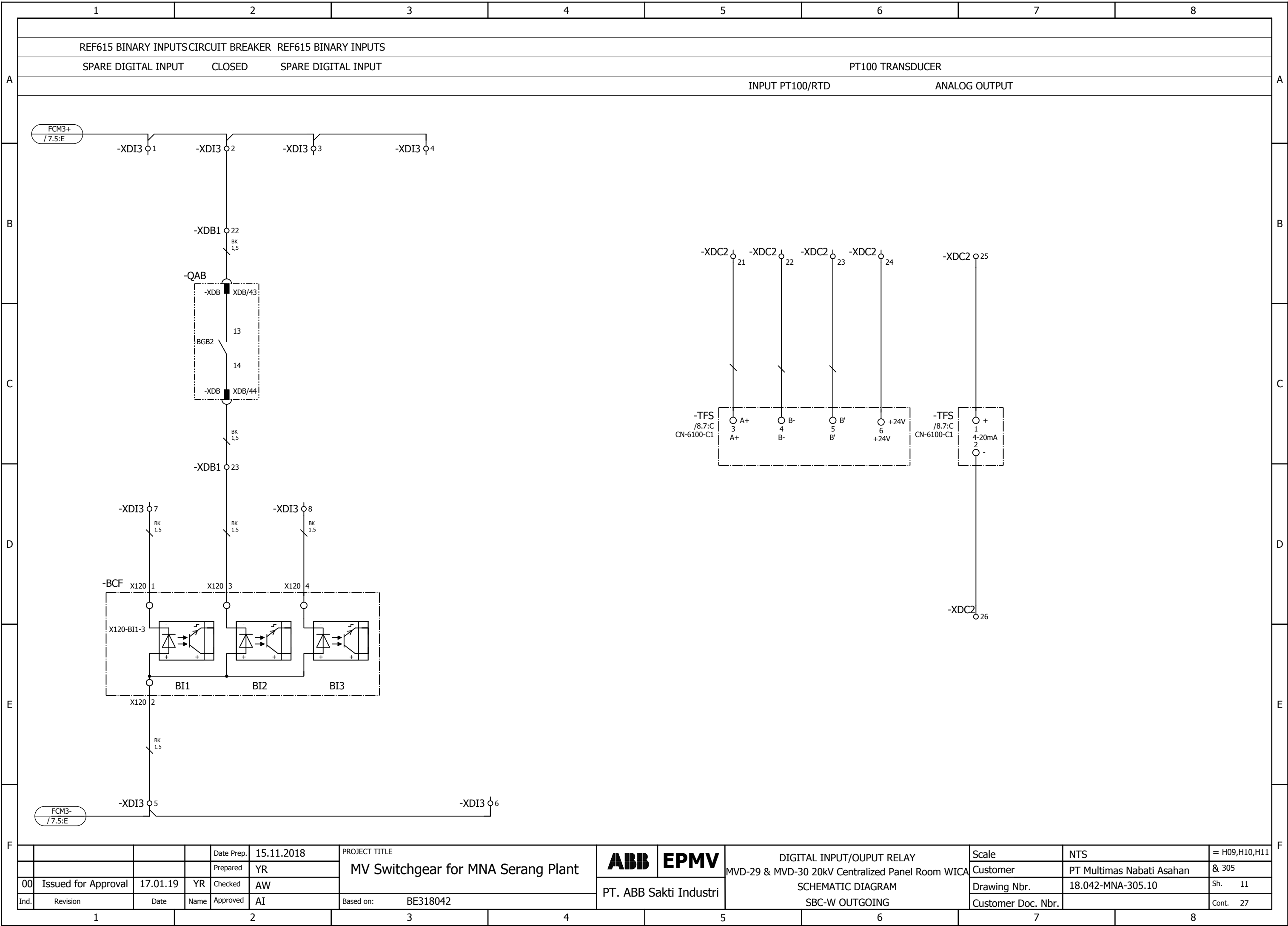
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	CIRCUIT BREAKER CONTROL MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale	NTS	= H09,H10,H11
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 305
00	Issued for Approval	17.01.19	YR	Checked	AW					Drawing Nbr.	18.042-MNA-305.10	Sh. 9
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri		SCHEMATIC DIAGRAM SBC-W OUTGOING	Customer Doc. Nbr.		Cont. 27

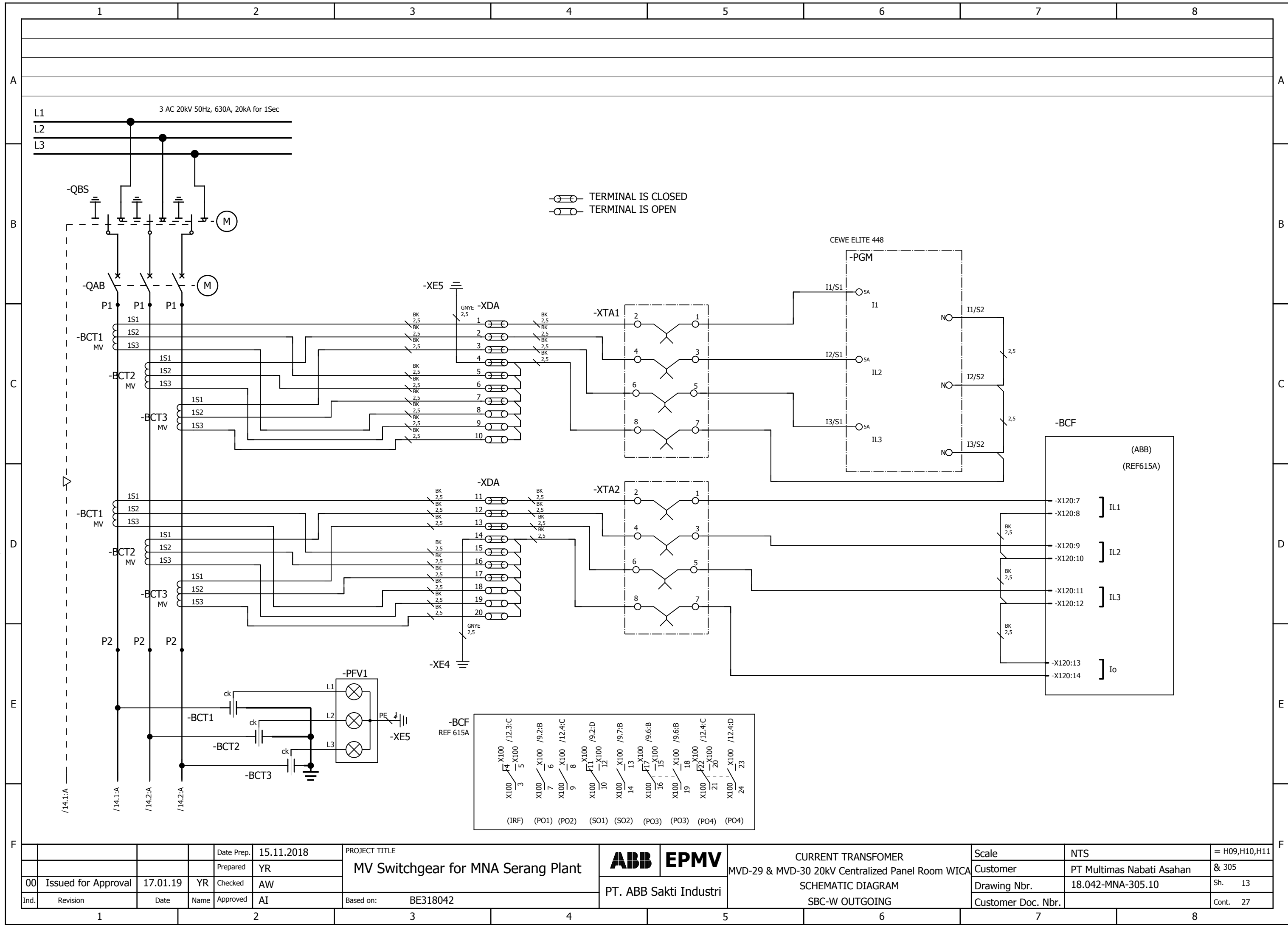
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant			AUXILIARY MOTOR & RELAY MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale	NTS	= H09,H10,H11
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 305
00	Issued for Approval	17.01.19	YR	Checked	AW		PT. ABB Sakti Industri	SCHEMATIC DIAGRAM		Drawing Nbr.	18.042-MNA-305.10	Sh. 10
Ind.	Revision	Date	Name	Approved	AI			Based on: BE318042		SBC-W OUTGOING	Customer Doc. Nbr.	

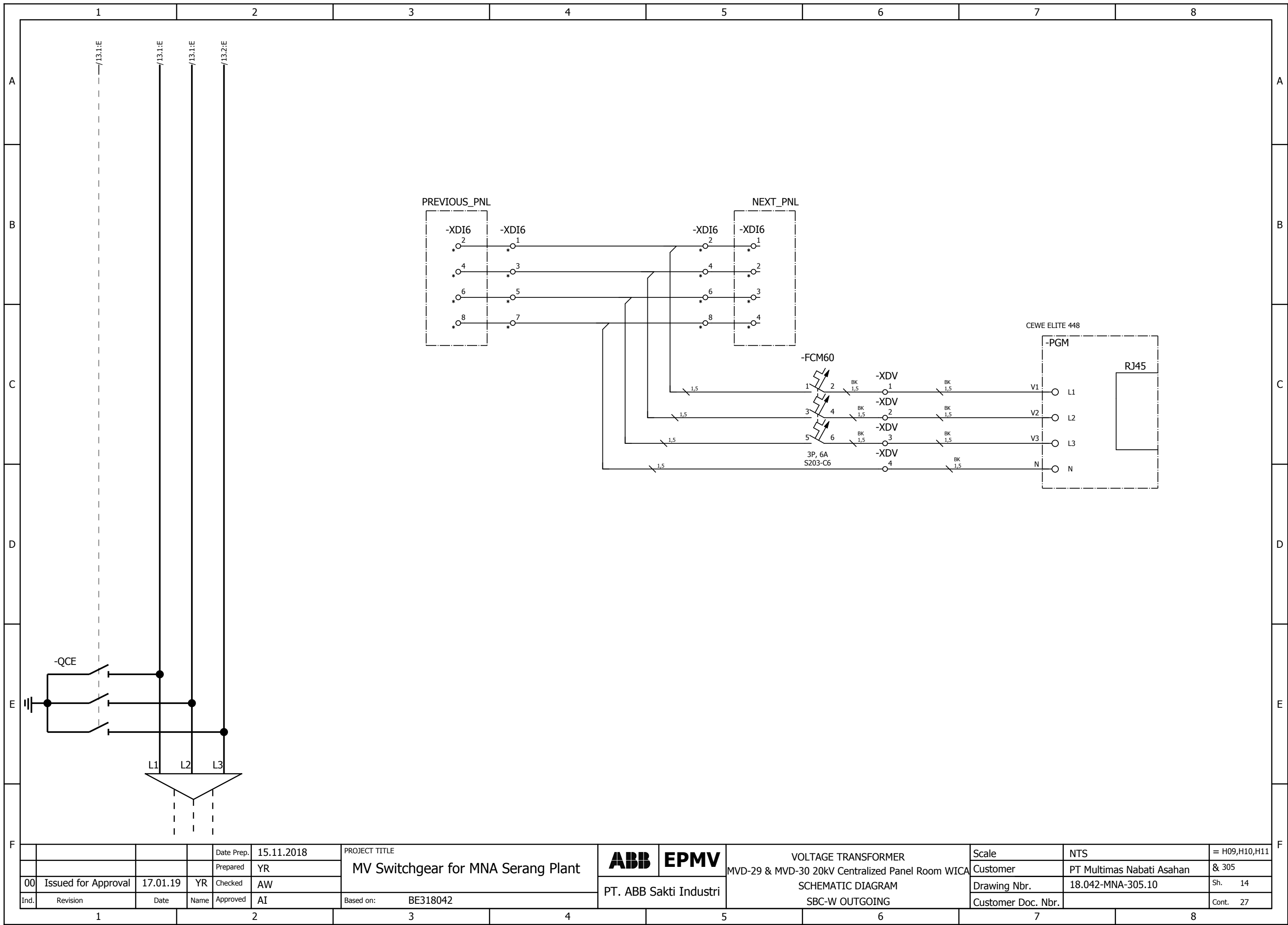
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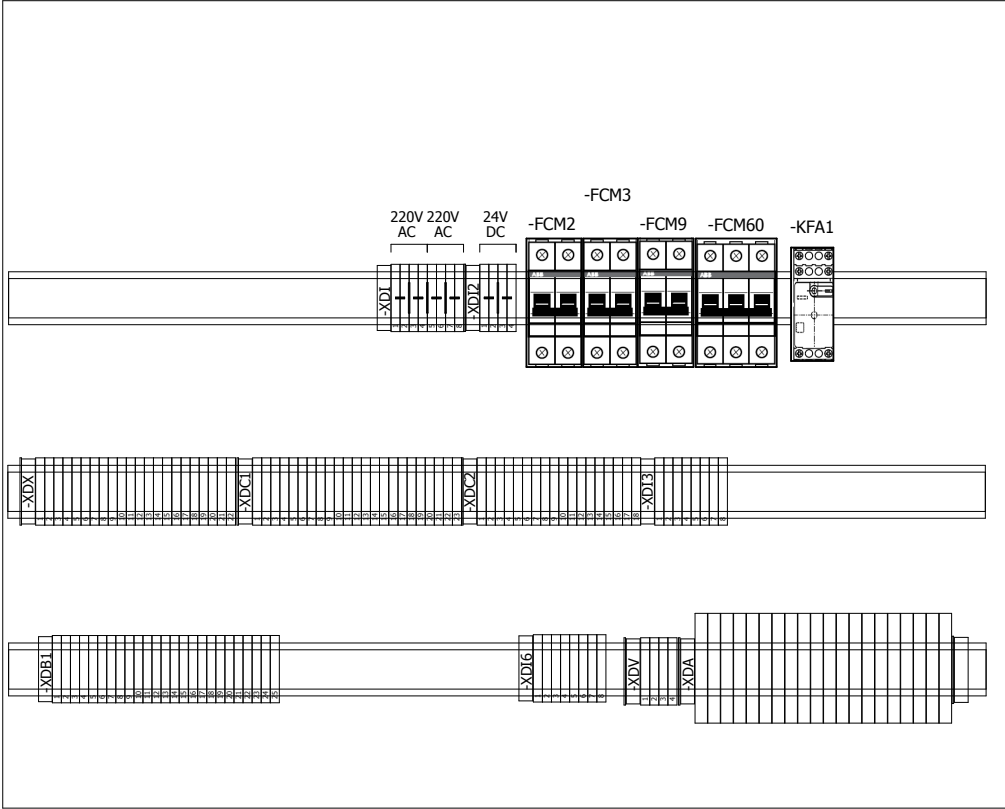
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	DIGITAL INPUT/OUPUT RELAY MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale	NTS	= H09,H10,H11
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 305
00	Issued for Approval	17.01.19	YR	Checked	AW		PT. ABB Sakti Industri	SCHEMATIC DIAGRAM		Drawing Nbr.	18.042-MNA-305.10	Sh.
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	SBC-W OUTGOING			Customer Doc. Nbr.		Cont. 27

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No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK

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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	LV BOX LAYOUT MVD-29 & MVD-30 20kV Centralized Panel Room WICA	Scale	NTS	= H09,H10,H11	
				Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan	& 305
00	Issued for Approval	17.01.19	YR	Checked	AW						SCHEMATIC DIAGRAM	Drawing Nbr.	18.042-MNA-305.10
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042			SBC-W OUTGOING	Customer Doc. Nbr.		Cont. 27	

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	1	2	3	4	5	6	7	8
A	Summarized parts list							
	IDABB_F02_005-IDABB							
	Carry over: 0.00							
	TAG ID		DESCRIPTION		MANUFACTUR/BRAND	PART NUMBER	QUANTITY	REMARKS
	-BCF		REF615A HBFAAAAACBC1ANA11G Configuration : A ; Power Supply : 48-250 Vdc; 100-240 Vac		ABB	HBFAAAAACBC1ANA11G	1	/8.2:C
	-EB		HEATER 230VAC, 50W		SANYANG	CDXN010033P0009	1	/7.6:E
	-FCM2		MCB S202-C10 MCB. 2P, 10A		ABB	2CDS252001R0104	1	/7.2:C
	-FCM3;-FCM9		MCB S202-C6 MCB. 2P, 6A		ABB	2CDS252001R0064	2	/7.5:C;/7.6:C
	-FCM60		MCB S203-C6 MCB. 3P, 6A		ABB	2CDS253001R0064	1	/14.6:C
	-KFA1		Auxiliary Relay 230AC 4COS		PHOENIX	1YRA3DL266230001	1	/9.7:E
	-PFG1		PILOT LIGHT GREEN #CL-523G#230VAC CL-523G		ABB	1SFA619402R5232	1	/8.4:E
	-PFR1		PILOT LIGHT RED #CL-523R#230VAC CL-523R		ABB	1SFA619402R5231	1	/8.4:E
	-PFY1		PILOT LIGHT YELLOW #CL-523Y#230VAC CL-523Y		ABB	1SFA619402R5233	1	/8.5:E
	-PGM1		POWER METER CEWE 448 Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC		CEWE	1YRA4EA0350001	1	/8.3:C
	-SFC		PUSH BUTTON, YW1B-M1E10-G GREEN COLOUR		IDEC	1YRA3DA14700001	1	/9.2:B
	-SFO		PUSH BUTTON, YW1B-M1E10-R RED COLOUR		IDEC	1YRA3DA14700002	1	/9.4:B
	-SFS		LOCAL/REMOTE Selector Switch Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position)		K&N	1YRA3DA17940011	1	/9.2:B
-TFS		Isolated PT100 Converter Model : CN-6100-C1		AUTONICS	CN-6100-C1	1	/8.7:C	
-XTA1;-XTA2		Current Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting		YONGSUNG	YSCTT-04C	2	/13.4:C;/13.4:D	

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[illegible]

[illegible]

[illegible]

[illegible]

F				E		D		C		B		A																			
Ind. Revision Date Name Approved				REMARK		CABLE		EQUIPMENT		Wire No		Terminal number		TYPE		WIRE NO.		EQUIPMENT		CABLE		REMARK									
				BK 1,5 mm ²				-PGM:V1						1		PT 2.5				-FCM60:2				BK 1,5 mm ²							
				BK 1,5 mm ²				-PGM:V2						2		PT 2.5				-FCM60:4				BK 1,5 mm ²							
				BK 1,5 mm ²				-PGM:V3						3		PT 2.5				-FCM60:6				BK 1,5 mm ²							
				BK 1,5 mm ²				-PGM:N						4		PT 2.5				-XD16:8				1,5 mm ²							
==MVD-29_MVD-30=H09,H10,H11+-XDV																															
REMARK: ☆ INTERPANEL LIGHT BUS,OFFERED TOGETHER WITH PANELS. △ OUTGOING CONTROL CABLE (Supply by Customer'). ● BRIDGE BAR. ✕ PARTITION PLATE. (TEST JUMPER.																															
				Date Prep.		15.11.2018		PROJECT TITLE				ABB		EPMV		Terminal diagram				Scale				= H09,H10,H11							
				Prepared		YR		MV Switchgear for MNA Serang Plant				PT. ABB Sakti Industri						Customer		PT Multimas Nabati Asahan		& 305									
				Checked		AW												Drawing Nbr.				Sh. 26									
Ind.				Revision		Date		Name		Approved		AI		Based on:		BE318042		Customer Doc. Nbr.						Cont. 27							
1				2				3				4				5				6				7				8			

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